Data exploration

Q1. Check for missing values in the 'Price' column. How would you handle products with missing price information?

Product ID	Product Brand	Quantity	Category	Price (cleaned)
07-JUN-UK	Sony Headphones	45	Electronics	Unknown
14-AUG-US	Coleman Camping Tent	10	Outdoor	Unknown
03-JUN-CA	Ray-Ban Sunglasses	25	Fashion	Unknown

Q2. If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.

Product ID	Product Brand	Category (cleaned)	Suggested category
22-MAY-US	North Face Backpack	Unknown	Outdoor / Accessories (confirm)
25-NOV-AU	Adidas Sneakers	Unknown	Fashion
08-DEC-DE	Nespresso Coffee Maker	Unknown	Kitchen
19-JUL-BR	Xiaomi Fitness Tracker	Unknown	Electronics

Q3. Identify any typos present in the "Category" column.

Product ID	Product Brand	Raw category	Cleaned category
07-JUN-UK	Sony Headphones	Electroni	Electronics
18-FEB-CA	Fitbit Smartwatch	Electroni	Electronics
16-APR-ES	Bose Headphones	Electroni	Electronics
20-AUG-CN	Huawei Smartwatch	Electroni	Electronics

Q4. Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.

Product ID	Column	Before	After (cleaned)
28-JAN-US	Product Name	laptop	Laptop
11-APR-US	Product Name	smartphone	Smartphone
19-JUL-UK	Product Name	T-shirt	T-Shirt
16-APR-ES	Product Name	headphones	Headphones
27-JAN-IT	Product Name	laptop	Laptop
07-JUN-UK	Category	Electroni	Electronics
18-FEB-CA	Category	Electroni	Electronics
16-APR-ES	Category	Electroni	Electronics
20-AUG-CN	Category	Electroni	Electronics

Q5. Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

Product ID	Product	Brand	Price	Qty	Category	Found at rows	Removed
17-JUN-IN	Laptop	HP	950	25	Electronics	12 and 34	Row 34
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	17 and 25	Row 25
16-APR-ES	Headphones	Bose	250	20	Electronics (Electroni)	16 and 29	Row 29

Q6. Split the "Product ID" column into two separate columns for "Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.

Product ID	Manufacturing Date	Country Code
28-JAN-US	28-JAN-2026	US
15-FEB-US	15-FEB-2026	US
03-MAR-US	03-MAR-2026	US
11-APR-US	11-APR-2026	US
22-MAY-US	22-MAY-2026	US
07-JUN-UK	07-JUN-2026	UK
19-JUL-UK	19-JUL-2026	UK
23-AUG-UK	23-AUG-2026	UK
05-SEP-UK	05-SEP-2026	UK
14-OCT-UK	14-OCT-2026	UK
17-JUN-IN	17-JUN-2026	IN
25-NOV-AU	25-NOV-2026	AU
08-DEC-DE	08-DEC-2026	DE
18-FEB-CA	18-FEB-2026	CA
16-APR-ES	16-APR-2026	ES
21-AUG-CA	21-AUG-2026	CA

Product ID	Manufacturing Date	Country Code
20-AUG-CN	20-AUG-2026	CN
27-JAN-IT	27-JAN-2026	IT
01-MAR-UK	01-MAR-2026	UK
14-AUG-US	14-AUG-2026	US
14-MAY-RU	14-MAY-2026	RU
09-JAN-CA	09-JAN-2026	CA
19-JUL-BR	19-JUL-2026	BR
29-SEP-CA	29-SEP-2026	CA
03-JUN-CA	03-JUN-2026	CA
11-JUL-CA	11-JUL-2026	CA
07-MAR-CA	07-MAR-2026	CA
13-APR-CA	13-APR-2026	CA
24-MAY-CA	24-MAY-2026	CA
02-DEC-CA	02-DEC-2026	CA
09-JUL-FR	09-JUL-2026	FR

Q7. Merge the "Brand Name" and "Product Name" columns into one column named Product Brand.

Brand Name	Product Name	Product Brand
Dell	Laptop	Dell Laptop
Nike	Sneakers	Nike Sneakers
Keurig	Coffee Maker	Keurig Coffee Maker
Samsung	Smartphone	Samsung Smartphone
North Face	Backpack	North Face Backpack
Sony	Headphones	Sony Headphones
Adidas	T-Shirt	Adidas T-Shirt
Ninja	Blender	Ninja Blender
Apple	Tablet	Apple Tablet
Timberland	Hiking Boots	Timberland Hiking Boots
Hp	Laptop	Hp Laptop
Adidas	Sneakers	Adidas Sneakers
Nespresso	Coffee Maker	Nespresso Coffee Maker
Fitbit	Smartwatch	Fitbit Smartwatch
Bose	Headphones	Bose Headphones
Samsonite	Laptop Bag	Samsonite Laptop Bag

Brand Name	Product Name	Product Brand
Huawei	Smartwatch	Huawei Smartwatch
Asus	Laptop	Asus Laptop
Oakley	Sunglasses	Oakley Sunglasses
Coleman	Camping Tent	Coleman Camping Tent
Nikon	Camera	Nikon Camera
Panasonic	Microwave	Panasonic Microwave
Xiaomi	Fitness Tracker	Xiaomi Fitness Tracker
Google	Smartphone	Google Smartphone
Ray-Ban	Sunglasses	Ray-Ban Sunglasses
Vitamix	Blender	Vitamix Blender
Zara	Dress	Zara Dress
Hamilton	Toaster	Hamilton Toaster
Garmin	Fitness Tracker	Garmin Fitness Tracker
Levi'S	Jeans	Levi'S Jeans
Casio	Watch	Casio Watch

Q8. Format the data type of the "Price" column to a currency format.

Product Brand	Price (currency)
Dell Laptop	₹1,000.00
Nike Sneakers	₹80.00
Keurig Coffee Maker	₹130.00
Samsung Smartphone	₹900.00
North Face Backpack	₹70.00
Sony Headphones	Unknown
Adidas T-Shirt	₹30.00
Ninja Blender	₹90.00
Apple Tablet	₹500.00
Timberland Hiking Boots	₹130.00
Hp Laptop	₹950.00
Adidas Sneakers	₹90.00
Nespresso Coffee Maker	₹120.00
Fitbit Smartwatch	₹150.00
Bose Headphones	₹250.00
Samsonite Laptop Bag	₹50.00

Product Brand	Price (currency)
Huawei Smartwatch	₹160.00
Asus Laptop	₹980.00
Oakley Sunglasses	₹150.00
Coleman Camping Tent	Unknown
Nikon Camera	₹700.00
Panasonic Microwave	₹80.00
Xiaomi Fitness Tracker	₹150.00
Google Smartphone	₹800.00
Ray-Ban Sunglasses	Unknown
Vitamix Blender	₹400.00
Zara Dress	₹60.00
Hamilton Toaster	₹40.00
Garmin Fitness Tracker	₹130.00
Levi'S Jeans	₹50.00
Casio Watch	₹100.00

Q9. Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY" format.

Product ID	Current	DD-MM-YYYY
28-JAN-US	28-JAN-2026	28-01-2026
15-FEB-US	15-FEB-2026	15-02-2026
03-MAR-US	03-MAR-2026	03-03-2026
11-APR-US	11-APR-2026	11-04-2026
22-MAY-US	22-MAY-2026	22-05-2026
07-JUN-UK	07-JUN-2026	07-06-2026
19-JUL-UK	19-JUL-2026	19-07-2026
23-AUG-UK	23-AUG-2026	23-08-2026
05-SEP-UK	05-SEP-2026	05-09-2026
14-OCT-UK	14-OCT-2026	14-10-2026
17-JUN-IN	17-JUN-2026	17-06-2026
25-NOV-AU	25-NOV-2026	25-11-2026
08-DEC-DE	08-DEC-2026	08-12-2026
18-FEB-CA	18-FEB-2026	18-02-2026
16-APR-ES	16-APR-2026	16-04-2026
21-AUG-CA	21-AUG-2026	21-08-2026

Product ID	Current	DD-MM-YYYY
20-AUG-CN	20-AUG-2026	20-08-2026
27-JAN-IT	27-JAN-2026	27-01-2026
01-MAR-UK	01-MAR-2026	01-03-2026
14-AUG-US	14-AUG-2026	14-08-2026
14-MAY-RU	14-MAY-2026	14-05-2026
09-JAN-CA	09-JAN-2026	09-01-2026
19-JUL-BR	19-JUL-2026	19-07-2026
29-SEP-CA	29-SEP-2026	29-09-2026
03-JUN-CA	03-JUN-2026	03-06-2026
11-JUL-CA	11-JUL-2026	11-07-2026
07-MAR-CA	07-MAR-2026	07-03-2026
13-APR-CA	13-APR-2026	13-04-2026
24-MAY-CA	24-MAY-2026	24-05-2026
02-DEC-CA	02-DEC-2026	02-12-2026
09-JUL-FR	09-JUL-2026	09-07-2026

Q10. Apply data bar or color scales conditional formatting in the "Price" column.

Product Brand	Price
Dell Laptop	₹1,000.00
Nike Sneakers	₹80.00
Keurig Coffee Maker	₹130.00
Samsung Smartphone	₹900.00
North Face Backpack	₹70.00
Sony Headphones	Unknown
Adidas T-Shirt	₹30.00
Ninja Blender	₹90.00
Apple Tablet	₹500.00
Timberland Hiking Boots	₹130.00
Hp Laptop	₹950.00
Adidas Sneakers	₹90.00
Nespresso Coffee Maker	₹120.00
Fitbit Smartwatch	₹150.00
Bose Headphones	₹250.00
Samsonite Laptop Bag	₹50.00

Product Brand	Price
Huawei Smartwatch	₹160.00
Asus Laptop	₹980.00
Oakley Sunglasses	₹150.00
Coleman Camping Tent	Unknown
Nikon Camera	₹700.00
Panasonic Microwave	₹80.00
Xiaomi Fitness Tracker	₹150.00
Google Smartphone	₹800.00
Ray-Ban Sunglasses	Unknown
Vitamix Blender	₹400.00
Zara Dress	₹60.00
Hamilton Toaster	₹40.00
Garmin Fitness Tracker	₹130.00
Levi'S Jeans	₹50.00
Casio Watch	₹100.00

Q11. Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."

Product Brand	Category
Dell Laptop	Electronics
Nike Sneakers	Fashion
Keurig Coffee Maker	Kitchen
Samsung Smartphone	Electronics
North Face Backpack	Unknown
Sony Headphones	Electronics
Adidas T-Shirt	Fashion
Ninja Blender	Kitchen
Apple Tablet	Electronics
Timberland Hiking Boots	Outdoor
Hp Laptop	Electronics
Adidas Sneakers	Unknown
Nespresso Coffee Maker	Unknown
Fitbit Smartwatch	Electronics
Bose Headphones	Electronics
Samsonite Laptop Bag	Accessories

Product Brand	Category
Huawei Smartwatch	Electronics
Asus Laptop	Electronics
Oakley Sunglasses	Fashion
Coleman Camping Tent	Outdoor
Nikon Camera	Electronics
Panasonic Microwave	Kitchen
Xiaomi Fitness Tracker	Unknown
Google Smartphone	Electronics
Ray-Ban Sunglasses	Fashion
Vitamix Blender	Kitchen
Zara Dress	Fashion
Hamilton Toaster	Kitchen
Garmin Fitness Tracker	Electronics
Levi'S Jeans	Fashion
Casio Watch	Accessories